

How does sibling gender affect children’s academic performance: Data from Belgium (Flemish)

Group T - Aarush Bathula (5513578), Edward Chamberlain (5526786),
Jamie Chan (5534070), Will Murphy (2204888)

We examine whether sibling gender affects academic performance amongst Flemish students using HBSC survey data from years 2006, 2010 and 2014. We study individuals with one sibling and multiple siblings separately, estimating ordered probit models of self-reported academic achievement. Across each specification, marginal effects are small and statistically insignificant. Our findings indicate that sibling gender does not impact academic performance, suggesting educational gaps likely arise from factors outside sibship gender composition.

1 Introduction

Siblings influence children’s development through role modelling, social partners and family dynamics (McHale et al., 2014). Using HBSC data within the Belgium (Flemish) region, we estimate an ordered probit model to analyse whether the gender of a sibling impacts academic performance (University of Bergen, 2006, 2010, 2014). We hypothesise that having a same-gender sibling, particularly if are older, may improve a child’s performance as they might communicate and offer better support. Furthermore, siblings of the same gender may be more likely to study similar subjects and attend the same school, meaning that resources such as textbooks and tutoring can be shared between the siblings. Sibling gender can lead to fertility stopping behavior due to first son preference, albeit unlikely in Belgium (Clark, 2000). We find no statistically significant effects of sibling gender on performance.

This research question is important because policies such as mentor schemes for students who do not have a same gendered sibling may lead to better academic outcomes.

1.1 Literature Review

Shared environmental factors are core contributors to between-family variation in cognitive achievement (Chipuer et al, 1990; Hauser & Sewell, 1986). Sibships heavily populated with females encourages classroom excellence (Powell & Steelman, 1990). However, other research finds the opposite effect (Butcher & Case, 1994). More recent work shows that parental investments differ by child gender (Quadlin, 2019). Collectively, these studies show how sibling gender effects academic achievement through behavioural outcomes and resource allocation.

Other literature finds no effect of siblings gender on attainment, rather age spacing of siblings has a significant positive effect on attainment (Bu, 2016).

Overall, the literature provides mixed evidence for the sibling gender effect on academic performance. We aim to contribute to this literature by focusing on the Flemish region where socioeconomic segregation in schools is very high (Belfi, 2014).

2 Data

Health Behaviour in School-aged Children (HBSC) is a repeated cross-sectional dataset conducted by WHO, for adolescents aged 11, 13, 15 every four years. The Belgium (Flemish) sample contains 12,884 observations. We drop only children to focus on the sibling gender effect, leaving 10,889 observations. We create a "Single Sibling" sample for individuals with only one sibling leaving 4,481 observations. Restricting to two-child families allows us to isolate sibling gender effects without impeding additional family-size dynamics.

Our identification concern is that the sibling's gender is not randomly assigned. When parents fail to achieve their desired sex ratio, they revise their fertility decisions upward (Yamaguchi & Ferguson, 1995; Hank & Kohler, 2000).

Parents who have two children in this dataset, have most likely completed their fertility

progression once children have reached adolescence, mitigating gender bias in sibship composition.

2.1 Descriptive Statistics

Table 1 shows the descriptive statistics for "Single Sibling" and "All siblings" subsections of data. The total means and standard errors across both subsections are similar, providing validity for our single sibling analysis. Furthermore, the mean of "Same-sex sibling" for those with one sibling is approximately 0.5 which supports the quasi-random assignment of sibling gender assumption.

Table 2 shows the balance check of same-sex and opposite-sex sibling for "Single Sibling" group which compares the predetermined characteristics. For both "Same-sex" and "Opposite-sex" pairs, both sex groups are roughly similar, so treatment and control groups are nearly identical, reinforcing our quasi-random assignment assumption.

3 Empirical Analysis

3.1 Main Model & Identification Strategy

We estimate a parsimonious ordered probit model with the dependent variable: academic achievement $y_i \in \{1, 2, 3, 4\}$, the categories are: Below average, Average, Good, and Very good respectively.

The baseline group are individuals without a same sex sibling. We include control variables denoted by X_i .

$$P(y_i = j \mid x_i) = \Phi(\tau_j - \eta_i) - \Phi(\tau_{j-1} - \eta_i), \quad j = 1, 2, 3, 4, \quad (1)$$

$$x_i = (\text{Same-sex}_i, X_i)' \quad (2)$$

$$\eta_i = \beta_1 \text{Same-sex}_i + \beta_4' X_i \quad (3)$$

$-\infty \equiv \tau_0 < \tau_1 < \tau_2 < \tau_3 < \tau_4 \equiv +\infty$ are threshold parameters for academic performance

We cluster standard errors at the school level as pupils in the same institution share similar characteristics.

3.2 Controls

Our pre-determined control variables are Sex, Age, Income (proxied by affluence) and wave fixed effects.

- Females generally have higher academic achievement than males (Voyer & Voyer, 2014).
- Age is likely to influence educational maturity (Dhuey & Bedard, 2006).
- Income can influence educational value through a range of mechanisms (Brook-Gunn & Duncan, 1997). Income affect fertility decisions in developed countries however within the "Single sibling" dataset same sex sibling will be random (Hank, 2006).
- Wave fixed effects control for changes across years such as academic policies.

3.3 Results Discussion

Our main results are shown in Table 3 & 4. To test for how the effect of sibling gender differs across boys and girls, we computed the marginal effects for boys and girls at each achievement level. The conditional marginal effects are small in magnitude, as shown by Table 4, each below one percentage point, whilst being insignificant. This suggests sibling gender does not statistically influence academic performance for boys and girls.

3.4 Robustness Checks

Beyond the ordered probits, we include OLS and heterogeneity models. Table 5 indicates a slight increase in academic achievement at the age category of 13, with all other heterogeneity largely null. We also extend our analysis to multi-sibling households, only finding that girls might benefit from mixed-siblings more than boys of similar traits, as seen in Table 6. We further examine multi-sibling composition in Table 8.

We ran a regression solely for the 2014 data which had richer data, we included the additional control of immigrant origin and heterogeneity by teacher trust. Second-generation immigrant students tend to performed below natives (Azzolini et al, 2012). Despite this, the inclusion of these variables does not change the insignificant of sibling-gender effects. Tables 7 & 9 confirm null effects.

Using our largest standard clustered error of 0.0083, the Minimum detectable effect (MDE) at 10% significance level and 80% power was 2.32% points, meaning our study cannot detect our current estimates with all estimate lying below the MDE.

3.5 Limitations

We acknowledge that clustering at family levels further eliminates cultural and family environment differences across individuals to improve on the school clusters. Given that academic achievement is self-reported and prior work shows females over report, the measurement error in our dependent variable may be non-random and related to gender (Caskie et al, 2014). Perceptions of achievement may also be influenced by teacher trust (Terrier, 2020). Trust in teachers may suffer from reverse causality, reducing explanatory power. Hence, we rely on the assumption that the sibling gender is conditionally exogenous after restricting the sample.

4 Conclusion

Our findings reveal no statistically or economically meaningful effect of sibling-gender on academic performance for Flemish adolescents as all estimated marginal effects falling below the MDE.

Flanders should therefore target reducing socioeconomic gaps as these factors more likely drive achievement. Academic achievement is also self-reported, highlighting the need for providing objective performance through transcripts, to achieve consistent reporting across groups.

Future research could explore how birth order and resource allocation affect academic performance.

All members of the group contributed equally.

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6 Appendix

Table 1: Baseline Characteristics by Sibling Structure

	Panel A. One-sibling sample			Panel B. Multi-sibling sample (composition)			
	Total	Same-sex	Opposite-sex	Total	Only same-sex	Mixed siblings	Only opposite-sex
Academic achievement (1-4)	2.75 (0.83)	2.74 (0.83)	2.75 (0.82)	2.64 (0.85)	2.63 (0.85)	2.65 (0.85)	2.62 (0.83)
Same-sex sibling	0.49 (0.50)	1.00 (0.00)	0.00 (0.00)	0.81 (0.39)	1.00 (0.00)	1.00 (0.00)	0.00 (0.00)
Female	0.49 (0.50)	0.48 (0.50)	0.49 (0.50)	0.48 (0.50)	0.41 (0.49)	0.49 (0.50)	0.52 (0.50)
Age (years, 11/13/15 groups)	13.02 (1.67)	13.01 (1.67)	13.04 (1.66)	13.03 (1.66)	13.08 (1.62)	13.03 (1.68)	12.99 (1.62)
Family affluence (1-5)	2.84 (0.67)	2.85 (0.67)	2.83 (0.67)	2.87 (0.72)	2.85 (0.69)	2.88 (0.73)	2.87 (0.70)
Mother in main home (recoded)	1.03 (0.16)	1.02 (0.15)	1.03 (0.17)	1.05 (0.22)	1.06 (0.23)	1.05 (0.21)	1.05 (0.22)
Father in main home (recoded)	1.10 (0.30)	1.09 (0.29)	1.11 (0.32)	1.19 (0.39)	1.17 (0.37)	1.20 (0.40)	1.19 (0.39)
Number of siblings	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	3.20 (2.11)	2.44 (0.98)	3.67 (2.46)	2.41 (0.94)
Observations	4481	2212	2269	6408	1268	3945	1195

Notes: Entries are means with standard deviations in parentheses. Panel A restricted to children with exactly one sibling in the Flemish Belgium HBSC (2006, 2010, 2014). Panel B restricted to children with two or more siblings in the analysis sample. In Panel B: “Only same-sex” denotes children whose siblings are all the same sex as the child; “Mixed siblings” denotes at least one brother and at least one sister; “Only opposite-sex” denotes all siblings of the opposite sex. Estimates are weighted using HBSC sampling weights.

Table 2: Baseline Characteristics by Sibling Gender

	Same-sex (N=2212)	Opposite-sex (N=2269)	Difference
Female	0.48	0.49	0.01 (0.01)
Age	13.48	13.51	0.03 (0.05)
Family affluence (1–5)	2.85	2.83	-0.02 (0.02)
Mother in main home (recoded)	1.02	1.03	0.00 (0.00)
Father in main home (recoded)	1.09	1.11	0.02 (0.01)

Sample restricted to children with exactly one sibling in the Flemish Belgium HBSC (2006, 2010, 2014); total estimation sample size is $N = 4481$.

Entries are means for the same-sex sibling group (column 1) and the opposite-sex sibling group (column 2). Column 3 reports the difference in means (same-sex minus opposite-sex) with standard errors in parentheses.

Variables: female = 1 if the respondent is female; age = respondent's age in years; welloff = Family Affluence Scale (higher values indicate higher family socioeconomic status); motherhome1 and fatherhome1 = 1 if the mother/father lives in the same home.

The covariates reported in this table correspond to the predetermined controls included in the baseline regression specification.

Joint test of covariate balance from regression of same_sex on all covariates:

$F(9, 191) = 0.51$; p -value = 0.865.

Table 3: Effect of Same-Sex Sibling on Academic Achievement

	O.Probit	O.Probit	O.Probit	OLS
main				
Same-sex sibling	-0.015 (0.034)	-0.013 (0.032)	-0.013 (0.032)	-0.011 (0.024)
Observations	4,461	4,461	4,461	4,461
Log likelihood	-5399.339	-5258.008	-5257.087	-5340.919
Pseudo R ²	0.000	0.026	0.026	
R ²				0.062

Standard errors in parentheses.

Columns (1)–(3) estimate ordered probit models of the four-category academic achievement variable (1 = below average, 4 = very good).

Coefficients represent effects on the latent achievement index underlying the observed categories; negative values indicate lower predicted achievement.

Column (1) includes no additional controls; Column (2) adds predetermined controls (female, age group, family affluence); Column (3) further adds wave fixed effects. Column (4) reports the OLS analogue for ease of interpretation.

All models use survey weights and cluster standard errors at the school level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Marginal Effects of Same-Sex Sibling on Achievement Levels

	Below avg	Average	Good	Very good
	b/se	b/se	b/se	b/se
Same-sex sibling	0.0016 (0.0038)	0.0034 (0.0080)	-0.0014 (0.0035)	-0.0035 (0.0083)
Observations	4,461	4,461	4,461	4,461

Marginal effects from ordered probit (Table 3, column 3).

Shows change in probability of each achievement level from having a same-sex sibling.

Standard errors in parentheses, clustered at school level.

Table 5: Heterogeneity in Same-Sex Sibling Effect

	By sex	By age	By affluence
Academic achievement (1-4)			
Same-sex sibling	-0.053 (0.050)	-0.066 (0.053)	0.023 (0.146)
Same-sex $\times$ Female	0.082 (0.069)		
Same-sex $\times$ Age 13		0.169** (0.073)	
Same-sex $\times$ Age 15		0.002 (0.075)	
Same-sex $\times$ Affluence			-0.013 (0.050)
Observations	4,461	4,461	4,461
Log likelihood	-5256.288	-5254.230	-5258.932
Pseudo R ²	0.027	0.027	0.026

Standard errors in parentheses

Ordered probit with interactions. All models control for child sex, age, wave, and affluence.

Reference: Male (col 1), Age 11 (col 2). Column (3) uses linear interaction in affluence.

Standard errors in parentheses, clustered at school level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Sibling Composition Effects in Multi-Sibling Families

	O.Probit	OLS
main		
Boy: mixed siblings	-0.004 (0.047)	-0.005 (0.037)
Girl: mixed siblings	0.105*** (0.040)	0.080** (0.031)
Only same-sex siblings	0.038 (0.047)	0.028 (0.036)
Observations	6,386	6,386
Log likelihood	-7740.191	-7860.515
Pseudo R ²	0.019	
R ²		0.045

Standard errors in parentheses

Sample: children with 2+ siblings. Reference: only opposite-sex siblings.

Boy: mixed = boy with brother(s) and sister(s). Girl: mixed = girl with brother(s) and sister(s).

Only same-sex = all siblings same sex as respondent.

Controls: child sex, age group, wave, family affluence. SE clustered at school level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: 2014 Wave: Immigrant Origin and Teacher Trust Mechanisms

	O.Probit	OLS	By trust
main			
Same-sex sibling	-0.048 (0.053)	-0.035 (0.039)	-0.036 (0.065)
Immigrant origin	-0.092 (0.074)	-0.071 (0.055)	-0.046 (0.074)
Same-sex $\times$ Medium trust			0.053 (0.091)
Same-sex $\times$ High trust			-0.204 (0.219)
Observations	1,492	1,492	1,483
Log likelihood	-1755.583	-1789.395	-1711.115
Pseudo R ²	0.039		0.057
R ²		0.091	

Standard errors in parentheses

Sample: 2014 wave, single-sibling families only.

Immigrant origin: at least one parent foreign-born. Teacher trust: 3-level (low, medium, high; ref: low).

Controls: child sex, age, family affluence. SE clustered at school level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Composition Effects by Age

	Academic achievement (1-4)
Academic achievement (1-4)	
Boy: mixed siblings	0.065 (0.071)
Girl: mixed siblings	0.113** (0.054)
Age group 13.5 $\times$ Boy: mixed siblings	-0.049 (0.089)
Age group 15.5 $\times$ Boy: mixed siblings	-0.145* (0.085)
Age group 13.5 $\times$ Girl: mixed siblings	-0.018 (0.076)
Age group 15.5 $\times$ Girl: mixed siblings	-0.008 (0.068)
Only same-sex siblings	0.037 (0.047)
Observations	6,386
Log likelihood	-7737.949
Pseudo R ²	0.019

Standard errors in parentheses

Multi-sibling sample. Interactions with age groups. Reference age: 11.

Standard errors clustered at school level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: 2014 Full Sample Robustness Check

	O.Probit	OLS
main		
Same-sex sibling	-0.069*	-0.052
	(0.041)	(0.031)
Immigrant origin	-0.085	-0.065
	(0.058)	(0.044)
Observations	3,634	3,634
Log likelihood	-4341.712	-4416.075
Pseudo R ²	0.029	
R ²		0.070

Standard errors in parentheses

Sample: All families with 1+ siblings (2014 wave only).

Results subject to composition bias from parental fertility decisions.

Main analysis (Table 7) uses single-sibling sample.

Controls: child sex, age, family affluence. SE clustered at school.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$